

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-001: Student Input Field Validation for Add and Edit Operations

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation for Add and Edit Operations
Student Name	P.Tharanan
Student ID	JF/ICT/24/20
Implementation Status	Completed
Date	10 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators currently can enter student information without sufficient validation when adding or editing students. For example, a phone number containing only a few digits may be accepted, and names with fewer than three characters or weak passwords may also be entered.

As incorrect or insecure information can affect data quality and account security, the system should validate all student input fields before accepting the data.

The system should ensure that student names, registration number, email address, phone number, and password follow the required validation rules.

1.2 Existing Behaviour

Before implementation:

- StudentFormModal.jsx performs only basic required-field checking.
- A phone number with fewer than 10 digits may be accepted.
- A phone number with more than 10 digits may be entered.
- Non-numeric characters may be entered into the phone number field.
- First name and last name do not have a minimum 3-character requirement.
- Invalid characters may be entered into name fields.
- Email format is not sufficiently validated.
- Registration number does not have comprehensive input validation.
- New student passwords do not enforce a strong password policy.
- Backend validation does not comprehensively validate all student input fields.
- Invalid data may potentially be submitted directly to the backend without passing clientside validation.

1.3 Requested Change

A client-side search facility was required above the administrator's student table.

The search must:

- Validate all required student fields before submission.
- Require a valid registration number.
- Require first name and last name to contain at least 3 characters.
- Allow valid alphabetic characters in first name and last name.
- Reject invalid numbers or inappropriate special characters in names.
- Validate the email address using a valid email format.
- Require the phone number to contain exactly 10 digits.
- Allow only numeric characters in the phone number.
- Reject phone numbers containing fewer than 10 digits.
- Reject phone numbers containing more than 10 digits.
- Require a password when creating a new student.
- Require the password to contain at least 8 characters.
- Require at least one uppercase letter in the password.
- Require at least one lowercase letter in the password.
- Require at least one number in the password.
- Require at least one special character in the password.
- Display clear validation error messages to administrators.
- Prevent invalid data from being submitted from the frontend.
- Apply server-side validation to prevent client-side validation bypass.
- Preserve existing Add, Edit, and Delete functionality.
- Maintain existing administrator authentication and authorization.
- Continue securely hashing passwords before storing them in the database.

2. Implementation

2.1 Implementation Summary

The CR-004 requirements were implemented in the Validation.

Comprehensive input validation has been successfully implemented across both frontend and backend layers to ensure absolute data integrity. The system now strictly mandates a valid registration number, an exactly 10-digit numeric phone number, and properly formatted email addresses. Student first and last names are restricted to alphabetic characters only, requiring a minimum length of 3 characters while automatically blocking inappropriate numbers or symbols. Real-time validation error messages are now displayed to administrators to immediately prevent invalid submissions before they reach the server.

2.2 Affected Files

File	Change	Purpose
backend\src\controllers\studentController.js	Modified	Implemented strict frontend and backend validation for student fields, ensuring correct formatting for names, emails, 10-digit phone numbers, and robust 8-character passwords.

Additional files: None / [List any additional files if modified]

2.3 Technical Changes

Validation

A validation function was implemented to:

- Add new student, edit student form input fields.

Client-Side Validation

The validation logic checks the following student fields:

- Registration number
- First name
- Last name
- Email
- Phone number
- Password (Add new student form only)

Performance

The validation operates improve database performance and security.

2.4 Implementation Logic

1. Trim input fields on the frontend and backend to eliminate leading and trailing whitespaces before evaluation.
2. Define strict regular expressions for names, phone numbers, and password complexity criteria.
3. Verify minimum length requirements ensuring both first and last names contain at least 3 characters.
4. Enforce alphabetic boundaries on names using character checks to reject any numeric inputs or inappropriate special characters.
5. Validate phone number length to ensure it contains exactly 10 digits, blocking entries with fewer or more digits.
6. Restrict phone number content to numeric values only, completely rejecting alphabetic letters or symbols.
7. Evaluate password strength constraints to require at least 8 characters consisting of one uppercase letter, one lowercase letter, one number, and one special character for new entries.
8. Intermediary error propagation to immediately return a 400 Bad Request status code and a descriptive JSON error message to the administrator upon any validation failure.
9. **Preserve system state** by keeping original database ID references intact, ensuring uninterrupted Add, Edit, Delete, and password hashing workflows.

2.5 Error and Data Handling

- Immediate validation rejection triggers an HTTP 400 Bad Request status code, halting data persistence instantly when any rule is breached.
- Descriptive JSON error payloads are returned to the frontend, allowing administrators to see clear, actionable validation error messages.
- Pre-validation trimming safely sanitizes user inputs by stripping leading and trailing whitespaces from names, phone numbers, and passwords.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend testing.

Testing Types:

- Validation testing
- UI testing
- Error / no-result testing

Environment: Development

Browser: [Chrome]**Test Role:**

Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	First name with fewer than 3 characters. And Last name with 3 or more valid characters.	Require first name and last name to contain at least 3 characters	Require first name and last name to contain at least 3 characters	Pass
TC02	First name with 3 or more valid characters , Last name with fewer than 3 characters, Valid email format, Phone number with exactly 10 digits. Valid strong password.	Take input field value	Take input field value	Pass
TC03	Invalid characters in first name, Invalid characters in last name	Alphabetic characters only allowed in first name and last name	Alphabetic characters only allowed in first name and last name	Pass
TC04	Phone number with fewer than 10 digits, Phone number with more than 10 digits.	Phone number to contain exactly 10 digits	Phone number to contain exactly 10 digits	Pass

TC05	Non-numeric phone number	Alphabetic characters and symbols are not allowed in PhoneNumber	Alphabetic characters and symbols are not allowed in PhoneNumber	Pass
TC06	Password shorter than 8 characters., Password without uppercase letters, Password without lowercase letters., Password without numbers., Password without special characters.	Password must be at least 8 characters long and contain at least one uppercase letter, one lowercase letter, one number, and one special character.	Password must be at least 8 characters long and contain at least one uppercase letter, one lowercase letter, one number, and one special character.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	6
Passed	6
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Administrators cannot submit the form when required fields are empty	UI testing	Met
AC-002	First names containing fewer than 3 characters are rejected.	UI testing	Met
AC-003	Last names containing fewer than 3 characters are rejected.	UI testing	Met
AC-004	Names containing invalid numbers or inappropriate special characters are rejected.	UI testing	Met
AC-005	Invalid email addresses are rejected.	UI testing	Met

AC-006	Phone numbers containing fewer than 10 digits are rejected.	UI testing	Met
AC-007	Phone numbers containing more than 10 digits are rejected.	UI testing	Met
AC-008	Phone numbers containing non-numeric characters are rejected.	UI testing	Met
AC-009	A valid 10-digit phone number is accepted.	UI testing	Met
AC-010	Weak passwords are rejected when creating a new student.	UI testing	Met
AC-011	A password containing at least 8 characters, uppercase, lowercase, number, and special character is accepted.	UI testing	Met

5. Evidence

A maximum of **two relevant screenshots** should be attached.

Evidence 01 – Field required

Related: AC-001

The screenshot shows a web form titled "Add New Student" with a subtitle "Create a new student account." and a close button (X). The form contains six input fields arranged in two columns: "Registration Number", "First Name", "Last Name", "Email", "Phone Number", and "Password". A red exclamation mark icon and a tooltip message "Please fill out this field." are positioned over the "Registration Number" field. At the bottom right, there are two buttons: "Cancel" and "Create Student".

Figure 1: Add new student fields required.

Evidence 02 – All validation are pass

Related: AC-009, AC-011

Add New Student

Create a new student account.

Registration Number

REG003

First Name

Sarah

Last Name

Fernando

Email

admin@acc.ucj

Phone Number

1234567891

Password

.....

Cancel

Create Student

Students

Manage student accounts and profiles.

+ Add Student

Search by registration number, name, or email...

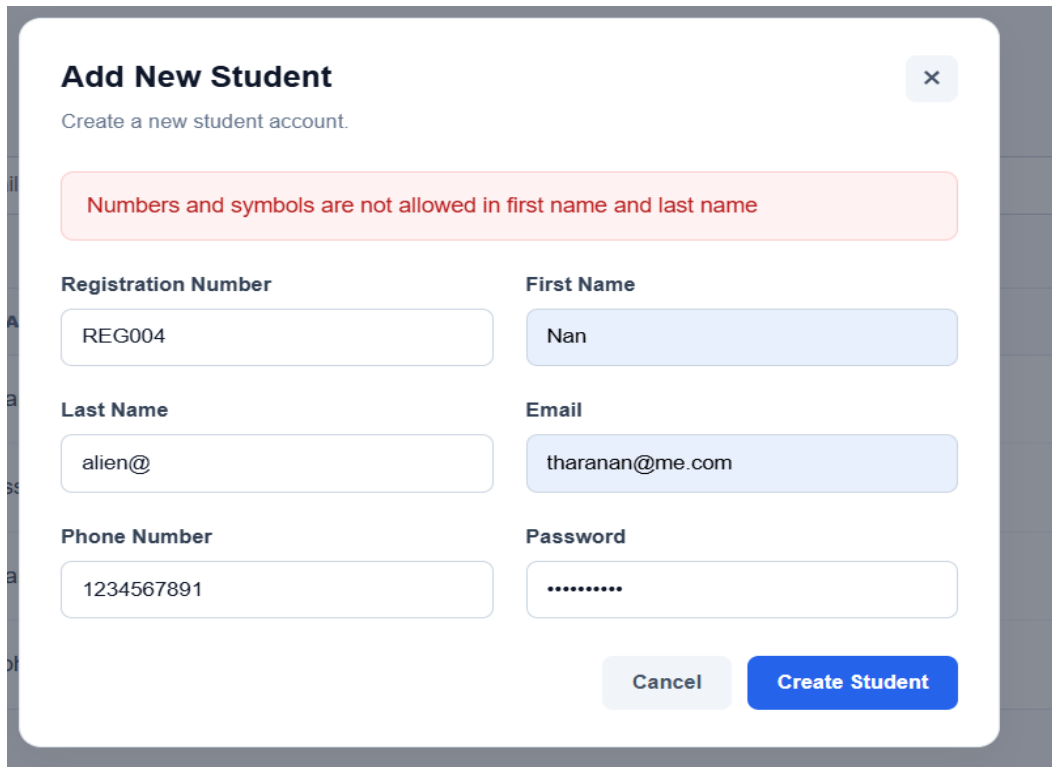
4 students

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
REG003	Sarah Fernando	admin@acc.ucj	1234567891	EditDelete
asq	ass aaa	admin@qw.co	1234567811a	EditDelete
REG002	Sarah Fernando	sarah@example.com	0723456789	EditDelete
REG001	John Perera	john@example.com	0712345678	EditDelete

Figure 2: All validation apply.

Evidence 03 – Last name validation

Related: AC-004



The screenshot shows a web form titled "Add New Student" with a subtitle "Create a new student account." A red error message at the top states: "Numbers and symbols are not allowed in first name and last name". The form contains six input fields: "Registration Number" (value: REG004), "First Name" (value: Nan), "Last Name" (value: alien@), "Email" (value: tharanan@me.com), "Phone Number" (value: 1234567891), and "Password" (value: masked with dots). At the bottom right are "Cancel" and "Create Student" buttons.

Figure 3: Last name has symbol.

6. Final Result

Item	Result
Requirements	Met / Partially Met / Not Met
Implementation	Completed / Partially Completed
Testing	Passed / Partially Passed / Failed
Acceptance Criteria	All Met / Partially Met / Not Met
Existing CRUD	Verified / Not Verified
Final Status	Accepted / Pending / Rejected

6.1 Final Summary

CR-004 was implemented as a comprehensive input validation and data security enhancement for the student management system by enforcing strict field-specific regular expressions on both client-side and server-side layers [CR-004]. The system now mandates valid registration numbers, standard email formats, and exact 10-digit numeric phone numbers while systematically rejecting invalid lengths, characters, or symbols [CR-004]. Student names are strictly restricted to valid alphabetic characters with a minimum length of 3 characters, and new accounts require an 8-character password featuring uppercase, lowercase, numeric, and special character combinations [CR-004]. Any input validation breach immediately blocks data submission on the frontend or halts persistence on the backend by throwing an HTTP 400 Bad Request error with explicit, descriptive JSON payloads for administrators [CR-004]. Crucially, this robust multi-tier validation architecture neutralizes client-side bypass attempts while completely preserving existing password hashing, administrator authorization, and legacy Add, Edit, and Delete workflows using the original student identifiers [CR-004].

6.2 Issues / Limitations

[State any remaining issues or limitations.]

If none:

No significant issues identified.

Student Name: P.Tharanan

Student ID: JF/ICT/24/20

Date: 10 October 2026

Signature: P.Tharanan

~ END OF REPORT ~